

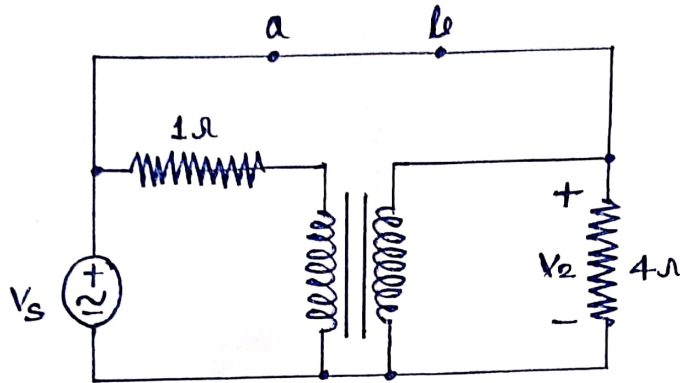
CTD QUIZ-4- SOLUTION (SET-A)

Sol (1)÷ Given that-

$$V_s = 117 \sin(500t) \text{ V}$$

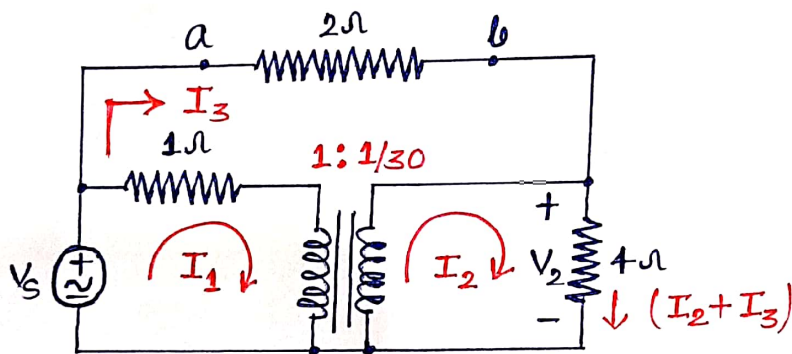
Turn ratio, $N_1:N_2 = 30:1$

(a)



Here, $V_2 = V_s = 117 \sin(500t) \text{ V} \rightarrow (2 \text{ Marks})$

(b)



$$I_2 = \frac{V_2 - V_s}{2} + \frac{V_2}{4} \quad \text{--- (1)} \rightarrow (1 \text{ Marks})$$

$$\frac{I_2}{I_1} = \frac{1}{a} = 30 \Rightarrow I_2 = 30 I_1 \rightarrow (1 \text{ Marks})$$

$$\frac{V_2}{V_1} = a = \frac{1}{30} \Rightarrow 30 V_2 = V_1 \rightarrow (1 \text{ Marks})$$

$$\frac{V_s - V_1}{1} = I_1 \rightarrow (1 \text{ Marks})$$

$$V_s - 30 V_2 = I_2 / 30 \quad \text{--- (2)}$$

$$4 I_2 = 2 V_2 - 2 V_s + V_2 \quad \text{--- (a)}$$

$$V_s - 30 V_2 = I_2 / 30 \quad \text{--- (b)}$$

By solving eqⁿ (a) & (b), we get -

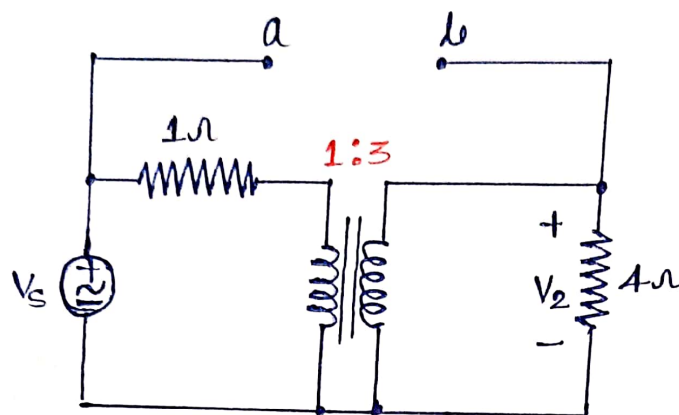
$$V_2 = 3.96 \sin(500t) \text{ V}$$

→ (4 Marks)

CTD QUIZ-4 SOLUTION (SET-B)

Sol(1)÷ Given that- $V_s = 720 \cos(120\pi t) \text{ V}$
Turn ratio, $N_1:N_2 = 1:3$

(a)



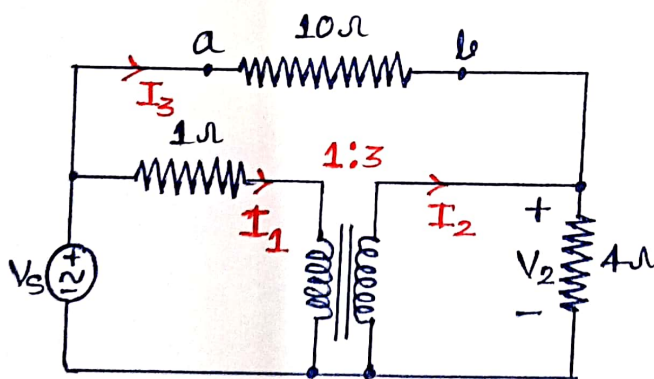
$$Z_{in} = \frac{Z_L}{a^2} = \frac{4}{3^2} = \frac{4}{9} \Omega$$

$$V_1 = \frac{(4/9)}{(4/9)+1} \cdot V_s = \frac{4}{13} V_s$$

$$V_2 = 3V_1 = 3 \times \frac{4}{13} V_s = \frac{12}{13} V_s = 664.6 \cos(120\pi t) \text{ V}$$

→ (3 Marks)

(b)



$$I_2 = \frac{V_2}{4} + \frac{V_2 - V_s}{10} \quad \text{--- (1)} \quad \rightarrow (1 \text{ Marks})$$

$$\frac{I_2}{I_1} = \frac{1}{3} \Rightarrow 3I_2 = I_1 \quad \rightarrow (1 \text{ Marks})$$

$$\frac{V_2}{V_1} = 3 \Rightarrow V_2 = 3V_1 \quad \rightarrow (1 \text{ Marks})$$

$$I_1 = \frac{V_s - V_1}{1} \quad \rightarrow (1 \text{ Marks})$$

$$V_s - V_1 = 3I_2 \quad \text{--- (2)}$$

By solving eqⁿ(1) & eqⁿ(2), we get -

$$V_2 = 676.62 \cos(120\pi t) \text{ V} \rightarrow (3 \text{ Marks})$$